

Introduction to Integers

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The set of **integers** consists of the **negative whole numbers**, **zero**, and the **positive whole numbers**.

Symbol for Integers

The symbol for the set of integers is $\mathbb{Z}$.

$\mathbb{Z}$

represents the set of all integers.

Set of Integers

The set of integers can be written as

$$\mathbf{Z} = \{\dots, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, \dots\}.$$

The integers continue indefinitely in both directions.

Numbers Not Included

Integers do **not** include **fractions**, **decimals**, or **irrational numbers**.

Examples that are not integers:

$$\frac{7}{2},$$

$$5.8,$$

$$\sqrt{2}$$

Whole Numbers and Integers

Every **whole number** is an integer.

However, not every integer is a whole number because integers also include negative numbers.

Whole Numbers $\subset$ Integers

Examples

Example 1

Is 15 an integer?

Yes. 15 is a positive whole number.

15 is an integer

Example 2

Is -23 an integer?

Yes. Negative whole numbers are integers.

-23 is an integer

Example 3

Is $\frac{7}{2}$ an integer?

No. It is a fraction.

$\frac{7}{2}$ is not an integer

Practice Exercises

Practice 1

Identify whether the number is an integer or not.

15

Practice 2

Identify whether the number is an integer or not.

$$-23$$

Practice 3

Identify whether the number is an integer or not.

0

Practice 4

Identify whether the number is an integer or not.

$$\frac{7}{2}$$

Practice 5

Identify whether the number is an integer or not.

5.8

Practice 6

Identify whether the number is an integer or not.

$$-11$$

Practice 7

Identify whether the number is an integer or not.

$$-0.75$$

Practice 8

Identify whether the number is an integer or not.

100

Activities

Activity 1

Identify whether the number is an integer or not.

42

Activity 2

Identify whether the number is an integer or not.

$$-17$$

Activity 3

Identify whether the number is an integer or not.

0

Activity 4

Identify whether the number is an integer or not.

$$\frac{9}{5}$$

Activity 5

Identify whether the number is an integer or not.

8.25

Activity 6

Identify whether the number is an integer or not.

-30

Activity 7

Identify whether the number is an integer or not.

$$-1.4$$

Activity 8

Identify whether the number is an integer or not.

256

Thank you!